Artificial Intelligence and Data Science

CSA1008 – Software Engineering

Assignment Title: CO2_At2_Architecure Design

Case Study Title: Flood Early Warning System

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1. System Requirement Analysis

1.1 System Objectives

The Flood Early Warning System is designed to detect floods early and send warning alerts to people. It helps citizens, disaster management authorities, and rescue teams reduce the impact of floods.

Objectives

- Detect floods at an early stage.
- Monitor weather continuously.
- Predict flood risks using AI.
- Send instant alerts to citizens.
- Help authorities manage rescue operations.
- Provide safe evacuation guidance.

1.2 Functional Requirements

The system should:

- Allow users to register and log in.
- Collect rainfall and river water level data.
- Monitor weather conditions.
- Predict flood risks.
- Send SMS, Email, and Mobile App alerts.
- Display flood status.
- Show evacuation routes.
- Store historical flood data.
- Generate reports.

1.3 Non-Functional Requirements

- High Performance
- Security
- Reliability
- Scalability
- Availability
- Maintainability

- User Friendly Interface

1.4 Quality Attributes

Scalability

Supports many users and weather sensors.

Reliability

Provides continuous monitoring.

Availability

System works 24×7.

Performance

Processes data quickly.

Security

Protects user information.

Maintainability

Easy to update and improve.

2. Software Architecture Selection

Selected Architecture

Hybrid Architecture

(Layered + Client-Server + Event-Driven)

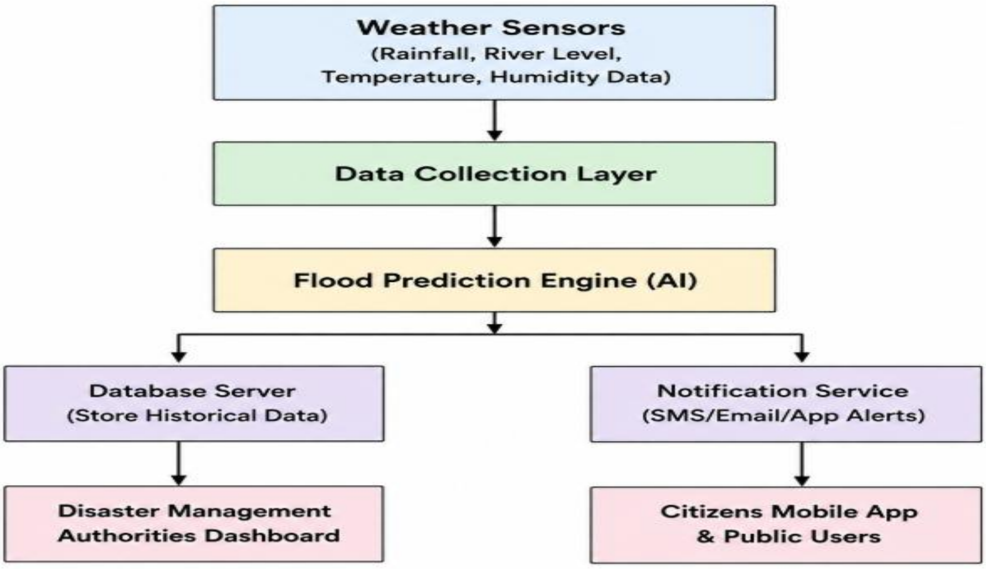
Why Hybrid?

The Flood Early Warning System has many functions like:

- Weather Monitoring
- AI Prediction
- Database
- Notifications
- Mobile App

A Hybrid Architecture is best because it combines the strengths of multiple architectures.

3. Software Architecture Diagram



4. Components and Their Interactions

1. Weather Sensors

Collect

- Rainfall
- Water Level
- Temperature
- Humidity

Send data continuously.

2. Data Collection Layer

Receives sensor data.

Checks whether the data is correct.

Forwards data to AI.

3. Flood Prediction Engine

Uses Machine Learning.

Analyzes weather conditions.

Predicts whether flooding will happen.

4. Database

Stores

- Sensor Data
- Flood History
- User Information
- Alert Records

5. Notification Service

Generates alerts.

Sends notifications through

- SMS
- Email
- Mobile App

6. Disaster Management Dashboard

Shows

- Live Flood Status
- Flood Prediction
- Emergency Alerts

Authorities monitor everything from this dashboard.

7. Citizens Mobile Application

Citizens can

- View flood status
- Receive alerts
- View evacuation routes
- Contact emergency services

System Workflow

1. Weather sensors collect weather data.
2. Data is sent to the Flood Prediction Engine.

3. AI predicts flood risk.
4. Data is stored in the database.
5. Notification service sends alerts.
6. Authorities monitor the situation.
7. Citizens receive warning messages.

5. Quality Attribute Analysis

Scalability

Supports thousands of sensors and users.

Security

- Login Authentication
- Encrypted Communication
- Secure Database

Performance

Processes sensor data quickly.

Sends alerts within seconds.

Reliability

Works continuously.

Stores backup data.

Maintainability

Modules can be updated separately.

Easy to add new features.

Availability

Cloud servers keep the system available 24×7.

6. Architectural Justification

Why Hybrid Architecture?

Hybrid Architecture combines

- Layered Architecture
- Client-Server Architecture
- Event-Driven Architecture

This makes the system

- Faster
- More secure
- Easier to maintain
- More reliable
- Easy to expand

Trade-offs

Advantages

- Easy maintenance
- Better scalability
- High reliability
- Fast notifications
- Better security

Disadvantages

- More complex to develop
- Higher implementation cost
- Requires cloud infrastructure

Future Enhancements

The architecture supports future improvements such as:

- Better AI models
- Drone monitoring
- Satellite image integration
- WhatsApp alerts

- Voice assistant support
- GIS-based flood mapping
- Multi-language support
- Offline emergency mode

Conclusion

The **Flood Early Warning System** uses a **Hybrid Architecture** to combine weather sensors, AI-based flood prediction, cloud storage, notification services, and mobile applications. This architecture provides fast flood detection, real-time alerts, secure data management, and effective communication between citizens and disaster management authorities. It is scalable, reliable, secure, and suitable for future enhancements, making it an ideal solution for improving disaster preparedness and public safety.

This report is structured to match the six sections required in your CO2 AT2 assignment: system requirements, architecture selection, architecture diagram, component interactions, quality attributes, and architectural justification.